No Easy Wins: A Contamination-Controlled Benchmark for Evaluating Anomaly Detection and Model Selection

Alexander Apartsin[a], Yehudit Aperstein[b,*]

[a]School of Computer Science, Faculty of Sciences, Holon Institute of Technology (HIT), Holon, Israel
[b]Intelligent Systems, Afeka Academic College of Engineering, Tel-Aviv, Israel

Abstract

Generic anomaly detection methods are commonly evaluated on benchmarks that combine datasets containing normal observations and labeled anomalies. Designing such benchmarks is itself challenging. Existing datasets may contain duplicated anomalies, train/test leakage, anomalies that are trivially separable, or anomalies that are effectively indistinguishable from normal observations. Some also provide insufficient data for reliable comparisons, while commonly used evaluation metrics can be distorted by differences in anomaly prevalence. Reliable benchmarking therefore requires curated datasets, prevalence-aware evaluation, and statistically grounded comparisons between methods.

We introduce ADReal, a benchmark constructed by systematically curating and filtering datasets from multiple existing anomaly detection benchmarks. We remove duplicates, contaminated anomalies, leakage, and datasets that are either trivially easy or effectively impossible, yielding 151 datasets spanning tabular and multivariate time-series data. On these datasets, we evaluate roughly thirty widely used detectors, including classical isolation-, density-, distribution-, and projection-based methods and deep one-class and reconstruction-based models. Detectors are evaluated using a prevalence-aware metric that enables more comparable performance estimates across datasets with different anomaly rates.

ADReal also addresses a second problem: model selection. In real applications, anomalies are rare and heterogeneous, so a representative labeled anomaly sample may not be available for model selection or hyperparameter tuning. The detector must therefore often be selected using normal data alone. ADReal provides a controlled testbed for evaluating such normal-only selection methods: candidate detectors are selected without access to true anomalies, while their eventual performance on those anomalies provides an independent measure of selection quality. Using this testbed we evaluate popular unsupervised model-selection heuristics and release their results alongside the benchmark and detector evaluations.

Finally, we introduce SPARC, a normal-only model-selection method that generates synthetic anomalies from normal data and ranks candidate detectors by their ability to identify them. SPARC significantly outperforms the strongest competing selection heuristics. Its regret, the performance gap relative to the best available detector, is 0.124, compared with 0.176 for entropy and 0.174 for mass-volume ($p < 0.001$ against each). The advantage is particularly large for time-series data, where regret decreases to 0.05 versus 0.15 for the strongest baseline, and also holds on tabular data. We release ADReal, the complete detector evaluations, SPARC, and the associated code.

1. Introduction

New anomaly detection methods are introduced and compared on benchmarks that pair normal observations with labeled anomalies, drawn from fraud, industrial monitoring, health, and cybersecurity. The ranking such a benchmark produces is only as trustworthy as the data behind it. And because no single detector dominates, Isolation Forest wins on some datasets, LOF on others, ECOD on yet others, a practitioner must still choose one detector per dataset, usually with no labeled anomalies to guide the choice. A decade of work has proposed label-free selection criteria for exactly this, from entropy and mass-volume curves [1] to internal relative-evaluation scores [2], meta-learning [3], and rank aggregation across a model pool [4]. Detectors and the selectors that choose among them are judged on the same benchmarks, so both inherit whatever those benchmarks get wrong.

Existing benchmarks are contaminated in ways that distort both kinds of comparison. Many anomalies are separable by a single feature or a one-line threshold rule, so every detector scores well and the choice between

*Corresponding author
Email address: apersteiny@afeka.ac.il (Yehudit Aperstein)

them barely matters. Some anomalies are exact copies of normal points, labeled both ways because the feature vector does not capture what makes them anomalous, and no detector can ever separate them. Normals leak between train and test when a dataset's duplicate rows are split at random, so a detector is scored on points it has already seen. Some tasks are effectively unsolvable, with no detector beating a random ranking, so a comparison on them carries no signal. Base rates drift so high that the scoring metric divides by a vanishing denominator. And differences are reported as point gaps with no test of significance. A benchmark built on such data measures artifacts, of the detectors and of the methods that choose among them.

We rebuild the benchmark and audit it until it holds. ADReal is a collection of tabular and multivariate time-series datasets produced by one detector-free pipeline whose construction rules each close one contamination channel. On the resulting 151 datasets we evaluate roughly thirty classical and deep detectors under a single consistent preprocessing, scored by a prevalence-aware metric, so no detector is handicapped by an inconsistency in the harness and the comparison is fair across datasets of different anomaly rates. ADReal then serves a second purpose: because a detector must in practice be chosen from normal data alone, the benchmark is also a controlled testbed for unsupervised model selection, where a method picks one detector without labels and is scored by its regret against the oracle-best detector, its criterion computed on held-out normals only. We evaluate eight popular selection methods under this protocol.

The honest protocol changes the result. Agreement-based selectors, which dominate under transductive evaluation, fall below a random pick once validation sees only normals; entropy and mass-volume survive. Building on this, we introduce SPARC, a structure-aware selector that grades detectors on synthetic probes built from the normal data, and it is the only method that significantly beats both survivors.

Our contributions:

1. **ADReal, a dataset and a protocol for evaluating anomaly detectors.** One detector-free, reproducible pipeline removes trivial, unsolvable, and mislabeled tasks, closes train/test leakage, caps the base rate, and deduplicates anomalies, so a detector's measured performance reflects skill rather than artifacts. From 173 constructed tasks, 151 survive.
2. **An evaluation of a rich set of modern detectors.** We fit roughly thirty classical and deep anomaly detectors under one consistent preprocessing, which gives each dataset its oracle-best detector.
3. **A protocol for evaluating unsupervised model selection, and an evaluation of the state of the art under it.** Beyond a detector's general performance, practice needs the best detector for a specific dataset given only its normal data, a different question. We score a selection method by its regret against the oracle-best detector, with its criterion computed on held-out normals only, and find that under this leakage-free protocol the agreement selectors (consensus, model-centrality, HITS) fall below a random pick while only entropy and mass-volume survive, reversing their apparent lead under transductive evaluation.
4. **SPARC, a model-selection method that beats the survivors.** SPARC builds synthetic anomalies from the normal data and grades detectors on how well they find them, and is the only method that significantly beats both entropy and mass-volume (regret 0.124; $p < 0.001$ against each), decisively on time series and holding on tabular data.

2. Related work

Anomaly detectors and benchmarks. Established detectors span the standard families: isolation-based (Isolation Forest [8]), density-based (LOF [9]), boundary-based (one-class SVM [10]), distribution-based (COPOD [11], ECOD [12], HBOS), projection-based (LODA [13], PCA), and deep methods (DeepSVDD [14] and later one-class and reconstruction models). Surveys [15] and empirical studies [16] document that no single detector dominates, which is what makes per-dataset selection necessary. ADBench [5] consolidates ODDS-derived tabular collections, TSB-AD [19] standardizes time-series detection, and the PyOD toolbox [6] makes large-scale comparison feasible; we build on all three. These resources standardize detectors and data, but not the protocol under which a *selection* method is judged, which is where contamination enters.

Unsupervised model selection. The central difficulty is choosing among detectors or configurations without labels. Internal metrics score a single detector from its own output: excess-mass and mass-volume curves [1] and the internal relative-evaluation score IREOS [2]. Rank-aggregation and meta-learning methods choose from a pool: MetaOD [3] regresses dataset meta-features to expected performance but needs a labeled meta-training corpus. A review of internal strategies [17] finds none reliably better than naive baselines. Most of these methods are evaluated *transductively*, computing their criterion on the same data, anomalies included, that supplies the ground truth. That protocol lets a selector's criterion see the outliers it is being asked to find. ADReal instead computes every selection criterion on held-out normals only, and the collapse we report in Section 6 is a direct consequence.

Consensus and outlier ensembles. Averaging or combining detector scores yields a consensus outlier score used both as an ensemble output and as an implicit selection signal that favors detectors agreeing with the majority [4, 18]; model-centrality and HITS-style rank aggregation formalize this. When validation contains no anomalies, the quantity these methods rank on is dominated by how detectors behave on normals, which is why they degenerate here. ADReal scores each selection method by regret against the oracle-best detector on held-out anomalies.

Contamination in existing benchmarks. Realism of the evaluation data has been raised before: synthetic-anomaly benchmarks depend heavily on how the anomalies are injected [7], and the DAMI study [16] documents how preprocessing, duplication, and normalization change outlier-detection results, and deduplicates its datasets for exactly the reasons we encounter. ADReal turns these scattered observations into a single detector-free construction protocol whose rules each remove one contamination channel, and whose target of measurement is the selection method rather than the detector.

3. The ADReal benchmark

ADReal is produced by a single pipeline that never consults an anomaly detector. Every filtering and splitting decision is made from the data alone, so no design choice can be tuned to flatter a selection method. The pipeline turns raw source datasets into evaluation tasks; each task provides normals for training, held-out normals for validation, and a test set of held-out normals plus hard anomalies. Every task clears the contamination controls below. Two of them (the separability floor and the selection-trivial floor) depend on detector scores and are therefore applied after the detector evaluation of Section 4; the rest are detector-free.

3.1. Data sources

ADReal draws from two modalities. Tabular tasks come from ADBench [5], DAMI [16], ODDBench, and OVRBench [20]; multivariate time-series tasks come from TSB-AD [19], windowed into fixed-length feature vectors. The pipeline constructs 173 tasks, of which 151 survive the filters below.

3.2. No trivially separable and no unsolvable anomalies

An anomaly a single feature or a one-line rule already isolates makes every detector look good and every selector's choice irrelevant. We remove such anomalies with a calibrated OR rule that consults no detector: each feature is thresholded by its own empirical-tail and interior-gap-histogram rarity at a 5% false-positive rate, and a point is flagged if any feature fires, with the same rule repeated in PCA-whitened space to catch oblique separations. For time series, a naive per-channel argmax fires on 56% of random label placements through multiple comparisons, so it is replaced by a permutation test against a circular-shift null at a 6.7% false-positive rate. At the opposite end, a task on which no detector beats a random ranking makes selection meaningless; a **separability floor** removes tasks whose oracle-best detector reaches base-rate-normalized average precision at or below 0.05, and a **selection-trivial floor** removes tasks on which every detector performs within a hair of the best, where any pick is a good pick.

3.3. No mislabeled anomalies and no leakage

Three failures let a selector be scored on the wrong answer. First, windowing mislabels: a sliding window inherits an anomaly flag from a single touched point, turning long stretches of normal signal into false anomalies. A stage-zero filter drops any series where fewer than half of anomaly points fall in contiguous segments of at least half a window. Second, label ambiguity: in coarse tabular data the same feature vector is sometimes labeled normal and sometimes anomalous, because the features do not capture what makes it anomalous; such an anomaly is an exact copy of a normal point and is unlabelable, so we remove every anomaly identical to a normal. Third, leakage: normals are split 60/20/20 into train, validation, and test by a **grouped split** that assigns each unique feature-vector, together with all its duplicate copies, entirely to one split, so no point ever appears in two splits while the density of repeated values is preserved within each split; for time series a contiguous-block split plus purging of boundary-straddling windows serves the same end. Every label-free selection criterion is computed on the validation split, which contains normals only, so no selector observes a test anomaly.

3.4. One consistent KPI with a bounded base rate

Every detector is scored by a single quantity, base-rate-normalized average precision, ap_norm = $(\mathrm{AP}-r)/(1-r)$ where r is the task's anomaly rate, comparable across tasks of different base rates. Because ap_norm divides by $1-r$, a task whose test set is mostly anomalies makes the metric degenerate; a **base-rate cap** holds the test anomaly fraction at or below one half by subsampling hard anomalies to parity with the held-out normals. A selection method is then evaluated by its regret, the ap_norm of the oracle-best detector minus the ap_norm of the detector it picked.

3.5. No duplicated anomalies and one significance test

Repeated near-identical anomalies inflate whichever detector happens to catch the template. Datasets are deduplicated across sources by name and dimensionality, and within a task the anomaly set is reduced to distinct cases. Selection methods are compared by a paired Wilcoxon signed-rank test on per-task regret, reported two-sided, together with the median and tail of the regret distribution. A raw difference of means that a handful of easy tasks could produce is never reported as a result.

4. Detectors and selection methods

Detector pool. The pool a selection method chooses from spans roughly thirty detectors. The classical part is the standard PyOD [6] set with fixed default hyperparameters: Isolation Forest, LOF, KNN, ECOD, COPOD, HBOS, PCA, CBLOF, LODA, and neighborhood-size and bin-count variants. The deep part adds one-class and reconstruction detectors trained per task: DeepSVDD [14], internal contrastive learning, reconstruction autoencoders and a variational autoencoder, and deep isolation. Every detector, classical and deep, is fit and scored on the same standardized features, so no detector is handicapped by a preprocessing inconsistency; the oracle-best detector per task defines the zero of regret.

Selection methods. We evaluate eight label-free selectors, all restricted to the leakage-free protocol (criteria on held-out normals only). *Entropy* (excess-mass) and *mass-volume* [1] score each detector by the shape of its score distribution over the data and a uniform reference sample. *Consensus* [4], *model-centrality*, and *HITS* rank detectors by agreement with the pool. *UDR* aggregates stability across configurations, *IFOREST-R* is a random-Isolation-Forest reference [17], and *IREOS* [2] scores the separability of the top-flagged points. Against these we place SPARC (Section 5). A uniform-random pick over the pool is the reference policy; a selector that fails to beat it carries no information.

5. SPARC: selection by synthetic probes

SPARC grades detectors on synthetic anomalies built from the normal data, then picks the detector that separates them best. It never observes a real anomaly.

Synthetic probes. From the validation normals, SPARC builds two kinds of synthetic anomaly. A *dependency* probe resamples a fraction of each point's features from their own marginal distributions, producing points whose per-feature values are ordinary but whose combination violates the normal correlation structure. A *marginal* probe pushes points into the tails of individual features. Each detector is graded by how well its scores separate each probe from the normals, giving a dependency grade and a marginal grade per detector.

Routing. SPARC routes between the best detector on each probe using a label-free property of the normal data: anisotropic datasets, whose covariance spectrum is concentrated in a few correlated directions, favor the dependency probe, while isotropic datasets favor the marginal probe; the two probe grades themselves supply a second, task-level signal of which probe is informative. Deep detectors tend to overfit the synthetic anomalies on coarse tabular data, separating the synthetic extremes without generalizing to the subtler real anomalies, so SPARC selects its probe winners among the classical detectors while the pool and the oracle still include the deep detectors.

6. Results

No single detector dominates, and deep does not beat classical. The detector evaluation is the substrate for every selection result: on each dataset it fixes the oracle-best detector that defines the zero of regret, and no detector holds that role everywhere. Figure 1 ranks all thirty-two detectors by mean base-rate-normalized average precision across the 151 datasets. Neighbor and density methods (LOF and KNN variants) take the top six places; the best deep detector, DeepSVDD, is individually strong but ranks seventh by mean. Classical detectors supply the per-dataset oracle on 118 of 151 datasets (78%) against 33 for the deep pool, and the best deep detector per task trails the best classical by 0.071 on average, so adding the entire deep pool to the classical one raises the oracle by only 0.021. Fairly evaluated under one consistent preprocessing, classical detectors carry this benchmark, and none of the selection results below depend on the deep detectors.

Agreement-based selectors fall below random under the leakage-free protocol. Figure 2 gives the selection regret of every method on the 151-task benchmark. The agreement-based methods that lead under transductive evaluation, consensus, model-centrality, and HITS, all land *above* the uniform-random reference (regret 0.232 to 0.235 versus 0.220), each significantly worse than SPARC ($p < 0.001$). With no anomalies in the validation split, the quantity these methods rank on is dominated by detector behavior on normals, and that quantity does not track detection skill. Entropy (EM) and mass-volume (MV) survive the protocol, at regret 0.176 and 0.174.

SPARC is the only selector that significantly beats both survivors. SPARC reaches regret 0.124, and the improvement over both survivors is significant at $p <$

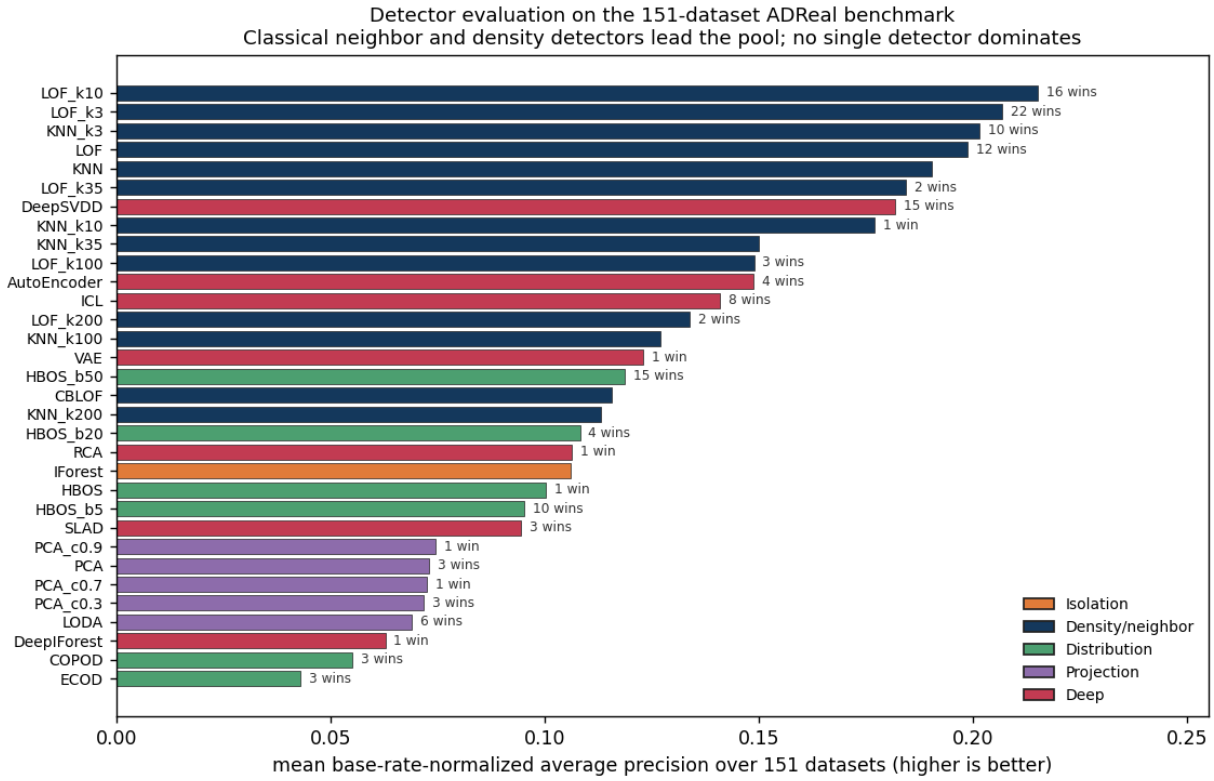

Figure 1. Detector evaluation on the 151-dataset ADReal benchmark: mean base-rate-normalized average precision per detector (higher is better), colored by family, with the number of datasets on which each detector is the oracle-best. Neighbor and density detectors lead; classical detectors supply the oracle on 78% of datasets, and the deep pool adds only 0.021 to the average oracle.

0.001 against each. No other method clears both. The margin does not depend on the deep detectors: as the detector evaluation above shows, a fairly preprocessed classical pool supplies almost every dataset's oracle, so the selection comparison does not hinge on the deep models.

The advantage is decisive on time series and holds on tabular data. Figure 3 breaks the result down by modality. On time-series window features SPARC reaches regret 0.054 against 0.155 for entropy and 0.169 for mass-volume, a threefold reduction ($p < 0.001$). On tabular data SPARC beats entropy on the ODDBench collection (0.151 versus 0.193, $p = 0.038$) and ties it on OVRBench (0.144 versus 0.167). The synthetic probes transfer cleanly to the structured window features of time series; on the coarse, low-cardinality features of the tabular semantic collections they are weaker proxies, and there entropy is already strong, so the two methods are matched.

7. Analysis and discussion

Where the synthetic probes work. SPARC's selection quality follows the fidelity of its probes to real anomalies, and that fidelity follows the feature representation. Time-series window features carry rich dependency and marginal structure, so the synthetic probes rank detectors as the real anomalies do and SPARC selects near the oracle. The tabular semantic collections encode each record in a handful of coarse features that under-determine the label, so a synthetic anomaly is only a loose proxy for a real one; there the probes rank detectors weakly and SPARC gains no edge over entropy, which is already a strong criterion on that data. The benchmark thus locates precisely where label-free structure-aware selection helps and where no method has an edge.

Why honest measurement matters. The collapse of agreement-based selectors is not a small correction. Under a transductive protocol these methods can appear to lead; under the leakage-free protocol they fall below a random pick. The two protocols disagree about which methods work, and only one of them measures label-

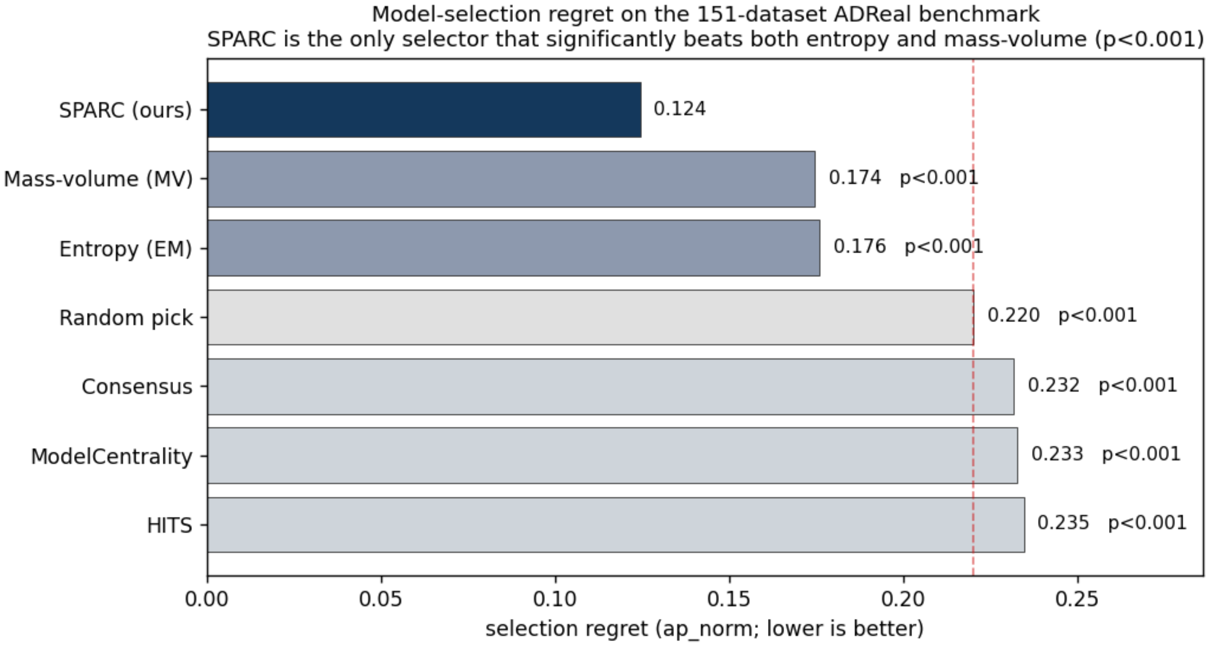

Figure 2. Model-selection regret on the 151-task ADReal benchmark (base-rate-normalized average precision; lower is better), with the two-sided paired Wilcoxon p-value against SPARC. Entropy and mass-volume survive the leakage-free protocol; the agreement-based selectors (consensus, model-centrality, HITS) fall at or above the random reference (dashed). SPARC is the only selector that significantly beats both survivors.

free selection as it is actually deployed. Constructing the benchmark also surfaced defects that quietly move results: anomalies identical to normal points, duplicate normals split across train and test, unbounded base rates, and a preprocessing inconsistency that handicapped classical detectors and inflated the apparent value of deep ones. Each was removed, and the corrected benchmark reports a selection landscape in which a structure-aware method that reads the geometry of the normal data leads the field.

8. Limitations

ADReal covers tabular and multivariate time-series data; univariate series and other modalities are out of scope here, and the construction rules would need modality-specific instantiation to extend. SPARC's advantage is decisive on time series and holds on tabular data, but on the coarse tabular semantic collections its synthetic probes are weak proxies and it only matches entropy; a probe construction that better fits low-cardinality features is the natural next step. The pool of detectors is finite, so the oracle that defines regret is the best *available* detector, not the best possible one.

9. Conclusion

Benchmarks for anomaly detection, and for the label-free methods that select among detectors, have been measuring artifacts: trivial and unsolvable tasks, anomalies that are copies of normals, leaked normals, degenerate base rates, and untested differences. ADReal rebuilds the measurement with a detector-free pipeline, audited until each of these channels is closed. On the clean benchmark it evaluates roughly thirty detectors under one consistent, prevalence-aware protocol, then evaluates selection methods by regret against the oracle-best detector under a leakage-free protocol. The honest protocol overturns the standing picture: agreement-based selectors fall below random, and SPARC, which grades detectors on label-free synthetic probes of the normal data, is the only selector that significantly beats both surviving baselines, decisively on time series and holding on tabular data.

Data and code availability

The benchmark construction pipeline, the per-task detector scores, the selection-method implementations, and the figures are released at the project repository. A permanent archive with a citable DOI accompanies the camera-ready version.

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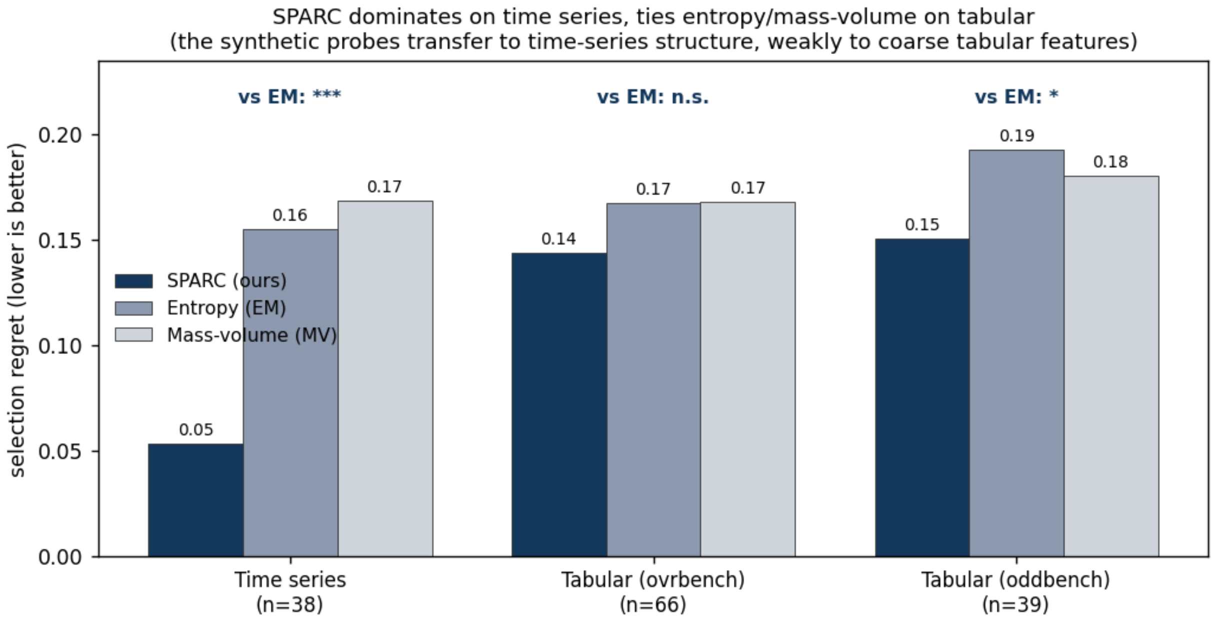

Figure 3. Selection regret by modality for SPARC versus the two surviving baselines. SPARC's advantage is decisive on time series ($p < 0.001$), significant on ODDBench ($p = 0.038$), and a tie on OVRBench. Significance is the paired Wilcoxon test of SPARC against entropy.

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